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MIL-C-50652/6A (AR)
26 November 1990
SUPERSEDING
MIL-C-50652/6 (MU)
15 April 1971

MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY: 10548122

(REGISTER A, RANGE)

This specification is approved for **use within the U.S. Army Armament, Munitions and Chemical Command, and Is available for use by all Departments and Agencies of the Department of Defense.**

1. SCOPE

1.1 Scope. This specification covers the detail requirements for one type of circuit card assembly. The requirements of this specification together with the general requirements of MIL-C-50652 comprise the complete requirements for the range register A used in Range Comuter CP-888/VPS-2, part of Radar Set AN/VPS-2 (see 6.1 of MIL-C-50652) .

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Benefical comments (recommendations, additions, deletions) and any pertinent data which may bc of use in improving this document be addressed to: Commdcr, U.S. Army ARDEC, ATTT: SMCAR-BACS, Peccatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

MILITARY

MIL-C-50652 - Circuit Board Assembly, General Specification for

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

U. S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

10548122 - Register A, Range

PACKAGING DATA SHEET

PS10548122 - Packaging of Circuit Card Assembly, Range Register A

(Copies of drawings, publications and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of conflict between the text of this specification and the references cited herein the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be In accordance with the applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 10548122, all associated drawings, and with all requirements specified in applicable specifications and standards. The circuit card assembly shall meet the requirements of MIL-C-50652.

3.3 Environmental service conditions. As specified in paragraph 3.3 of MIL-C-50652.

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3.4 Electrical. Each conductor shall be electrically continuous as specified on the schematic diagram of the circuit card assembly.

3.5 Performance. For the requirements of 305.1 through 3-5.3.4 the following conditions shall apply:

a. Logic 0: 0.0 ± 0.5 volt direct current (Vdc) level.

be Logic 1: ± 1.0 vdc level.

c m Positive pulse: a change in direct current (de) level from 0.0 ± 0.5 volt (V)-to ± 1.0 V and return to 0 V.

d. Negative pulse: a change in dc level from ± 3.5 ± 1.0 V to 0.0 ± 0.5 V and return to ± 3.5 V.

3.5.1 Input power. The input power applied between terminals P1-30, 31 and 32 (+5V and terminals Pl-24, 25 and 26 (+5 V return) shall be 5.o ± 0.3 Vdcj at 300 milliamperes (mA), regulated, with an alternating current (at) ripple less than 50 millivolts (mV) peak-to-peak.

3.5.2 Input signal. The input signal shall be a positive pulse of 20 kilohertz (KHz) and 0.8 ± 0.1 microsecond (us) duration.

3.5.3 Output signals.

3.5.3.1 Up count signals. With the input logic levels applied as specified in Table I, step 1, the output signals shall be symmetrical *square* waves for-the set and-reset-side, with frequencies as specified in Table II. Comparing two consecutive set or reset stage output signals, they shall be in phase, indicating a count up direction control.

TABLE 1. Input logic levels

	INPUT TERMINALS				
STEP	16	53	11	14	51
1	1	o	1	1	1
2	0	1	1	1	1
3	1	0	0	1	1

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TABLE II. Output signals

SET SIDE TERMINALS (TP)		RESET SIDE TERMINALS (TP)		SIGNAL FREQUENCY
7	1	13	21	10 KHz
9	2	12	20	5 KHz
56	3	10	19	2.5 KHz
37	4	21	18	1.25 KHz
33	5	19	17	625 Hz
35	6	23	16	312.5 Hz
30	7	36	15	156 Hz
40	8	22	14	78 Hz
48	9	45	13	39 Hz

3.5.3.2 Down count signals. With the input logic levels applied as specified in Table I, step 2, the output signals shall be symmetrical square waves for the set and reset side, with frequencies as specified in Table II. Comparing two consecutive set or reset stage output signals, they shall have a phase shift, indicating a count down direction control.

3.4.3.3 Reset signals. With the input logic levels applied as specified in Table I, step 3, the output signals shall be logic levels as specified in Table III.

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TABLE 111. Reset output signals

LOGIC 1 SET SIDE TERMINALS (TP)		LOGIC 0 RESET SIDE TERMINALS (TP)	
7	1	13	21
9	2	12	20
56	3	10	19
37	4	21	18
33	5	19	17
35	6	23	16
38	7	36	15
40	8	22	14
48	9	45	13

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3.5.3.4 Pulse output signals. With the input logic levels applied sequentially as specified in Table IV, the pulse output signals shall be as specified in Table IV.

TABLE IV. Pulse output signals

STEP	INPUT TERMINALS 16	53	11	14	51	OUTPUT TERMINAL	OUTPUT (TP)	SIGNAL CHARACTERISTIC 1
1	0	1	1	1	1	5	22	Positive pulse, 100 us PRF
2	1	0	0	1	1	5	22	Logic 0 level
3	0	1	1	1	1	1	23	Positive pulse, 200 us PRF
4	0	1	1	1	1	3	24	Positive pulse, 0.4 ms PRF
5	0	1	1	1	1	6	25	Positive pulse, 0.8 ms PRF
6	0	1	1	1	1	55	26	Positive pulse, 1.6 ms PRF
7	0	1	1	1	1	43	10	Negative pulse, 25.6 ms PRF
8	1	0	1	1	1	43	10	Logic 1 level
9	0	1	1	1	1	49	11	Positive pulse, 25.6 mS PRF
10	0	1	1	1	1	34	12	Logic 1 level
11	1	0	1	1	1	34	12	Negative pulse, 25.6 ms PRF
12	0	1	1	0	0	49	11	Positive pulse, 50 us PRF

1 All pulses shall have a duration of 0.8 ± 0.1 us.

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3.6 First article. As specified in paragraph 3.6 of MIL-C-50652.

3.7 Soldering. As specified in paragraph 3.7 of MIL-C-50652.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. As specified in paragraph 4.1 of MIL-C-50652.

4.2 Classification of inspections. As specified in paragraph 4.2 of MIL-C-50652.

4.3 First article inspection. As specified in paragraph 4.3 of MIL-C-50652 .

4.4 Quality conformance inspection. As specified in paragraph 4.4 of MIL-C-50652.

4.5 Special sampling. As specified in paragraph 4.5 of MIL-C-50652.

4.6 Inspection equipment. As specified in paragraph 4.6 of MIL-C-50652 and 4.6.2.3.

4.6.1 Government-furnished inspection equipment. As specified in paragraph 4.6.1 of MIL-C-50652.

4.6.2 Contractor-furnished inspection equipment.

4.6.2.1 Government design. As specified in paragraph 4.6.2.1 of MIL-C-50652.

4.6.2.2 Contractor design. As specified in paragraph 4.6.2.2 of MIL-C-50652.

4.6.2.3 Test equipment. In conjunction with 4.6.2.1 and 4.6.2.2. the test equipment listed in Table V (Test equipment list) shall be used in the performance of the applicable test methods and procedures as specified in 4.7.

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TABLE V. Test equipment list.

ITEM NUMBER		DESCRIPTION
1.	Power supply	5.0 +0.3 Vdc, regulated. ac ripple: less than so mV peak-to-peak Capacity: 300 mA.
2.	Pulse generator	20 KHz positive pulse; 3.0 ± 0.1 Vdc amplitude, 0.8 ± 0.1 us wide. Pulse rise and fall times less than 0.1 us .
3.	Oscilloscope	Dual channel. Rise time of scope plus measuring probe: 10 ns or less. Sweep period: 50 ns per division or greater. Input resistance: greater than 8 megohms. Input capacitance: less than 25 pF.
4.	Test fixture	As specified in Figure 1. See Table VI for test fixture parts list.

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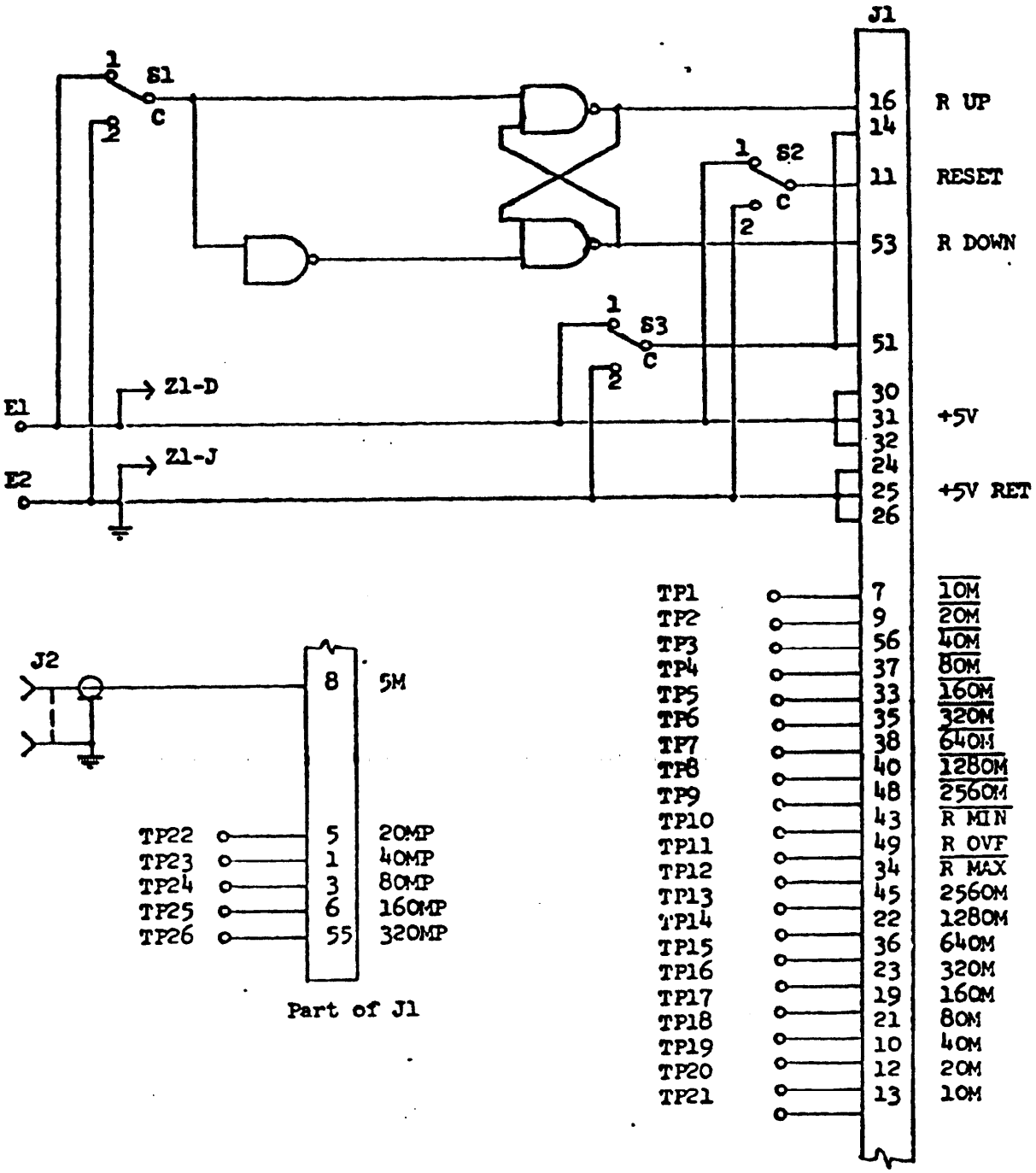


FIGURE 1. Test fixture setup.

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FIGURE 1. *Test* fixture setup.
TABLE VI. Test fixture parts list

DESIGNATION	NOMENCLATURE AND DESCRIPTION
E1	Binding post, H.H. Smith 257-R *or* equivalent.
E2	Binding post, H.H. Smith 257-B or equivalent.
J1	Connector; any of the following or equivalent. (1) 5893232-1, Malco Mfg. Col, Inc. Chicago, Illinois. (2) 500946-01, National Connector Corp., Minneapolis, Minn. (3) 07-5422-0198-01, Elco Corp., Willow Grove, Pa.
J2	Connector, receptacle, UG-25B/U or equivalent.
S1	Switch SPDT.
S2	Same as S1.
S3	Same as S1.
TP1 thru TP26	Tip jack, H.H. Smith 1501 or equivalent.
Z1	Integrated Circuit, Quad 2-Input NAND, SN5400 or equivalent.

4.7 Methods of inspection.

4.7.1 Environmental. As specified in paragraph 4.7.1 of MIL-C-50652.

4.7.2 Electrical. Continuity shall be inspected by standard measuring equipment (SME) and visual examination for compliance with 3.4.

4.7.3 Performance. The selected range registers shall be tested as follows: Connect range register A and test equipment (Table V, items 1 through 4) as shown in Figure 2. Pulses and logic levels shall be as specified in 3.5. Use switches S1 through S3 to apply logic levels as follows:

a. S1 - In position 1 applies logic 1 to P1-53 and logic 0 to P1-16, in position 2 applies logic 0 to P1-53 and logic 1 to P1-16.

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b. S2 - In position 1 applies logic 1 to P1-11 and in position 2 applies logic 0 to P1-11.

c. S3 - In position 1 applies logic 1 to P1-14 and 51, in position 2 applies logic 0 to P1-14 and 51.

Use oscilloscope (Table V, item 3) to observe input and output signals" as specified in 4.7.3.2 through 4.7.3.3.4.

4.7.3.1 Input power. Using power supply (Table V, item 1) apply input power as specified in 3.5.1.

4.7.3.2 Input signal. Using pulse generator (Table V, item 2) set up and apply input signal, as specified in 3.5.2, to 32 on test fixture (Table V, item 4).

4.7.3.3 Output signals.

4.7.3.3.1 UP count signals. Using switches S1 through S3 apply logic levels as specified in Table I, step I and observe the output signals as specified in Table II. Use oscilloscope channels A and B to compare set and reset output stages for phase as specified in 3.5.3.1.

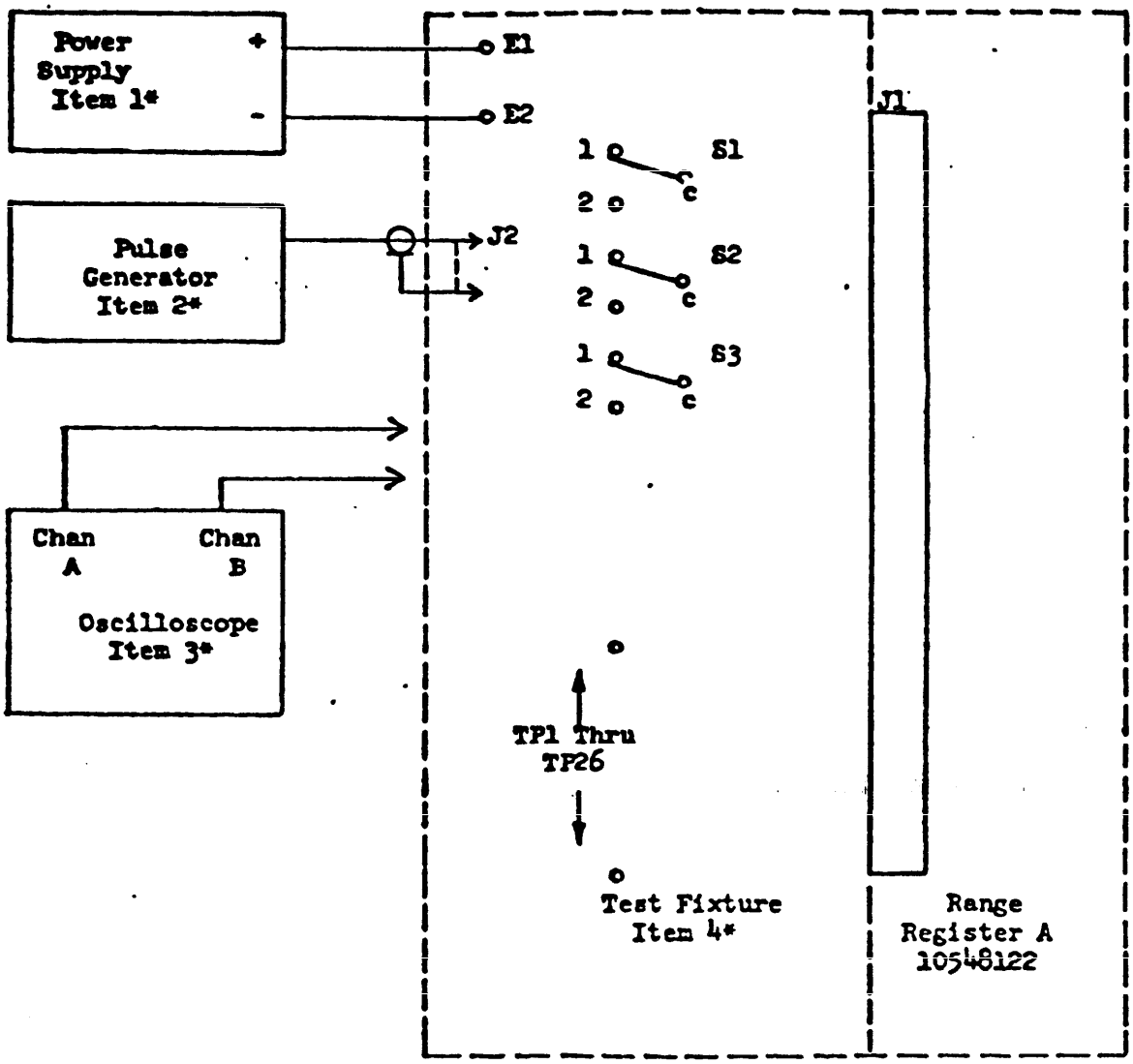
4.7.3.3.2 Down count signals. Using switches S1 through S3 apply logic levels as specified in Table I, step 2 and observe the output signals as specified in Table II. Use oscilloscope channels A and B to compare set and reset output stages for phase shift as specified in 3.5.3.2.

4.7.3.3 Reset signals. Using switches S1 through S3 apply logic levels as specified in Table I, step 3, and observe the output logic levels as specified in Table 111.

4.7.3.3.4 Pulse output signals. Using switches S1 through S3 apply logic levels as specified in Table IV and observe the output signals as specified in Table IV.

4.8 Soldering. Test as specified in paragraph 4.8 of MIL-C-50652.

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*Refer to 4.6.2.3

FIGURE 2. Range register A test setup.

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5. PACKAGING

5.1 General. Preservation, packing and marking shall be in accordance with Packaging Data Sheet PS10548122. The level of protection shall be as specified in the procurement document.

6. NOTES . As specified in section 6 Notes of MIL-C-50652.

Custodian:
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Preparing activity:
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-C-50652/6A (AR)	2. DOCUMENT DATE (YYMMDD) 26 November 1990

3. DOCUMENT TITLE
CIRCUIT CARD ASSEMBLY: 10548122 (REGISTER A, RANGE)

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER		
a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

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